

Simons Targeted Research Groups Grant Proposal

Prime Gap Physics: From Number Theory to the Standard Model

Principal Investigator: Jason Isaac Brodsky (California, 1976)
Framework: Arithmetic Physics — Targeted Investigation of Prime-Physics Connection
Request: \$300,000 over 3 years
Status: Rolling LOI

1. PROJECT SUMMARY

This targeted research group will investigate the connection between prime gap statistics and the Standard Model of particle physics. Building on our derivation of α from C_2 ($\alpha^{-1} = 2\pi/C_2 = 137.035999084\dots$), the group will extend this to all coupling constants and particle masses, develop the mathematical framework, and make testable predictions for precision experiments.

2. SCIENTIFIC BACKGROUND

The Prime-Electron Correspondence establishes a bijection between: - Prime gaps $d_n = p_{n+1} - p_n \leftrightarrow$ Proper time intervals $\Delta\tau_n = \kappa \cdot d_n$ - Twin prime constant $C_2 \leftrightarrow$ Fine structure constant $\alpha^{-1} = 2\pi/C_2$ - Record gaps $\{2,4,6,8,14\dots\} \leftrightarrow$ Lepton mass hierarchy - Riemann zeros $\gamma \leftrightarrow$ Worldline resonance frequencies - RH $\Leftrightarrow$ Worldline stability

Verified at 0.0019% precision (Reinman Numbers matching CODATA 2018).

3. RESEARCH OBJECTIVES

Year 1: Complete Coupling Derivations

1. **α_s from record gaps** — Strong coupling from $d_{\max} \sim \ln^2 x$ statistics
2. **G_F from modulo 6** — Weak coupling from gap parity classes
3. **Unification scale** — Gap convergence at directory 3.0 (RG flow)

Year 2: Mass Spectrum and Flavor

- 1. **Quark masses from gaps** — Gap modulo 3 → SU(3) color hierarchy
- 2. **CKM matrix from cross-correlations** — $V_{ij} \propto \sqrt{C_{ij}(k_*)}$
- 3. **Neutrino masses from gap asymmetry** — Sub-eV from gap statistics

Year 3: BSM and Experimental Interface

- 1. **Proton decay τ_p** — From gap stability: $\sim 10^{34}$ years (Hyper-K testable)
- 2. **Dark matter from missing gaps** — Direct detection signatures
- 3. **Gap fluctuation noise** — g-2, EDM, LZ/XENON predictions

4. TEAM (4 Researchers)

Researcher	Role	Expertise
PI (Brodsky)	Framework, computation, prime-electron correspondence	Arithmetic Physics, GPU computing
Co-PI 1 (Math)	Number theory, Hardy-Littlewood, explicit formulas	Analytic number theory, ζ -zeros
Co-PI 2 (Physics)	QFT, precision phenomenology, g-2, EDM	Particle physics, experimental tests
Co-PI 3 (Math Phys)	Holography, modular forms, TQFT	AdS/CFT, mathematical physics

5. BUDGET (\$300,000 over 3 years)

Category	Annual	Total
PI Summer Salary	\$15,000	\$45,000
Co-PI Summer Salaries (3 × \$10K)	\$30,000	\$90,000
Postdoc (shared)	\$60,000	\$180,000
Travel/Workshops	\$15,000	\$45,000
Computing	\$10,000	\$30,000
Indirect Costs	\$20,000	\$60,000
Total	\$150,000	\$450,000

Note: Requesting \$300K direct costs; indirect costs covered by institutions

6. QUALIFICATIONS

PI: 360+ document research program, computational verification, public data access.

Co-PIs: To be recruited from number theory (IAS, Princeton, Bonn), QFT (Fermilab, Harvard, Caltech), mathematical physics (Perimeter, IAS).

7. REFERENCES

All supporting materials in GrantProposals/ and CSMWip/
SubAtomicPrimeElectronCaldera/

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End of Simons Targeted Research Groups Proposal